

James B. Wyngaarden III

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ACADEMIC APPOINTMENTS

2026 Postdoctoral Fellow
Icahn School of Medicine at Mount Sinai—New York, NY
Department of Psychiatry
Advisor: *Muhammad A. Parvaz, Ph.D.*

EDUCATION

2026 Ph.D. Psychology (Cognition & Neuroscience)
Temple University—Philadelphia, PA
Department of Psychology & Neuroscience
Advisor: *David V. Smith, Ph.D.*
Dissertation: *Examining the effects of theta-frequency transcranial alternating current stimulation over left dorsolateral prefrontal cortex on reward-based learning across the adult lifespan*

2020 M.A. Experimental Psychology
Appalachian State University—Boone, NC
Department of Psychology
Advisor: *Andrew Monroe, Ph.D.*

2016 B.S. Psychology
University of North Carolina at Chapel Hill—Chapel Hill, NC
Department of Psychology & Neuroscience
Advisor: *Charlotte A. Boettiger, Ph.D.*

GRANTS, HONORS, & AWARDS

2025 Scientific Research Network on Decision Neuroscience and Aging (SRNDNA) Pilot Grant, NIH R24-AG076847-04, "Modulating reward-based learning in older adults using transcranial alternating current stimulation" (\$24,000)

2024 Ruth L. Kirschstein National Research Service Award (NRSA) Predoctoral Fellowship, National Institute on Aging (NIA), F31-AG085934, "Characterizing the role of confirmation bias in financial exploitation of older adults" (3 years, \$106,980)

2019 Three Minute Thesis (3MT) Competition—Appalachian State University, 1st Place (\$600)

2019 Wiley F. Smith Endowment (WiSE) Research Award (\$300)

2019 Student and Faculty Excellence (SAFE) Fund Research Grant (\$1,000)

PUBLICATIONS

Wyngaarden, J.B., III*, Nambiar, A.*, Dennison, J.B., Alloy, L.B., Fareri, D.S., Jarcho, J.M., & Smith, D.V. (2026) Trait reward sensitivity and behavioral motivation shape connectivity between the default mode network and the striatum during reward anticipation. *Cogn Affect Behav Neurosci* **26**, 838–849. <https://doi.org/10.3758/s13415-025-01396-9>.

- Wyngaarden, J.B., III**, Giovannetti, T., Krekelberg, B., Smith, D.V. (2026). Maladaptive Decision-Making in Older Adults: Confirmation Bias and Financial Exploitation. In: Smith, D.V., Lockwood, P.L., Fareri, D.S. (eds) *Neuroeconomics: Core Topics and Current Directions*. Springer, Cham. https://doi.org/10.1007/978-3-032-02925-6_30.
- Smith, D. V., **Wyngaarden, J. B., III**, Weinstein, S. M., Illenberger, N., Liu, Y., Siegel, J., & Krekelberg, B. (2026). α -tACS modulates reward-dependent pupil responses and corticostriatal connectivity. *Under review*. Preprint available at <https://doi.org/10.64898/2026.01.30.702963>.
- Smith, D. V., Zaff, O., **Wyngaarden, J. B., III**, Dennison, J. B., Sazhin, D., Chein, J., ... & Fareri, D. S. (2026). Social Closeness and Reward Sensitivity Enhance Corticostriatal Function during Experiences of Shared Rewards. *Biological Psychology*, 109209. <https://doi.org/10.1016/j.biopsycho.2026.109209>.
- Dwamena, D., Stanley, J. M., **Wyngaarden, J. B., III**, Kos, M., Sharp, C., Yang, Y., Jarcho, J., Fareri, D., & Smith, D. (2025). Neural and behavioral responses to social norm violations across trust and fairness contexts. *Under review*. Preprint available at <https://doi.org/10.1101/2025.11.20.689349>.
- Shah, S. F. A., Dennison, J. B., **Wyngaarden, J. B., III**, Clithero, J. A., & Smith, D. V. (2025, October 2). Ambiguity Preferences Are Distinct from Beliefs About Risk. *Under review*. Preprint available at https://osf.io/preprints/psyarxiv/wrej8_v1.
- Yang, Y., Ludwig, R.M., **Wyngaarden, J.B., III**, Dwamena, D., Hackett, K., Giovannetti, T., Fareri, D., Smith, D.V. (2025). Older adults show altered network connectivity during fairness decisions with similar and dissimilar partners. *Under Review*. Preprint available at <https://www.biorxiv.org/content/10.1101/2025.08.13.670194v1>.
- Wyngaarden, J.B., III***, Yang, Y.*, Dennison, J.B.*, Smith, D.V. (2025). A dataset of risky and ambiguous decisions using a novel linked colored lottery task across two datasets. *Data in Brief*. <https://doi.org/10.1016/j.dib.2025.111844>.
- Smith, D.V., Sharp, C.J., Dachs, A., **Wyngaarden, J.B., III**, Sazhin, D., Yang, Y., Kos, M., Tropea, T., Kohli, I., Clithero, J.A., Olson, I., Giovannetti, T., Fareri, D., Jarcho, J.M. (2024). Social reward and nonsocial reward processing across the adult lifespan: An interim multi-echo fMRI and diffusion dataset. *Data in Brief*. <https://doi.org/10.1016/j.dib.2024.110810>.
- Wyngaarden, J.B., III***, Johnston, C.R.*, Sazhin, D., Dennison, J.B., Zaff, O., Fareri, D., McCloskey, M., Alloy, L.B., Smith, D.V., & Jarcho, J.M. (2024). Corticostriatal responses to social reward are linked to trait reward sensitivity and subclinical substance use in young adults. *Social Cognitive and Affective Neuroscience*. <https://doi.org/10.1093/scan/nsae033>.
- Sazhin, D., **Wyngaarden, J.B., III**, Dennison, J.B., Zaff, O., Fareri, D., McCloskey, M.S., Alloy, L.B., Jarcho, J.M., & Smith, D.V. (2024). Trait reward sensitivity modulates connectivity with the temporoparietal junction and anterior insula during strategic decision making. *Biological Psychology*. <https://doi.org/10.1016/j.biopsycho.2024.108857>.
- Smith, D.V., **Wyngaarden, J.B., III**, Sharp, C.J., Sazhin, D., Zaff, O., Fareri, D., Jarcho, J.M. (2023). A neuroimaging dataset of social and nonsocial reward processing in young adults. *Data in Brief*. <https://doi.org/10.1016/j.dib.2024.110197>.

Jarcho, J.M., **Wyngaarden, J.B., III**, Johnston, C.J., Quarmley, M., Smith, D.V., & Cassidy, C.M. (2022). Substance abuse in emerging adults: The role of neuromelanin and ventral striatal response to social and monetary rewards. *Brain Sciences*, 12, 352.
<https://doi.org/10.3390/brainsci12030352>.

Monroe, A.E., **Wyngaarden, J.B., III**, & Plant, A.E. (2021). "They should have followed the rules": Trade-offs between fairness and authority values predict judgments of social justice protest. *Social Psychological and Personality Science*, 12(4), 471-478.
<https://doi.org/10.1177/1948550620923854>.

PEER REVIEW

- Cortex
- Frontiers in Human Neuroscience

Reviews completed in collaboration with senior mentor:

- Biological Psychiatry
- Human Brain Mapping
- Proceedings of the National Academy of Sciences—Nexus
- Social, Cognitive, and Affective Neuroscience

PRESENTATIONS

Wyngaarden III, J.B., Nambiar A., Dennison J.B., Alloy L.B., Fareri D.S., Jarcho J.M., Smith D.V. (October 2025). *Trait reward sensitivity and motivation shape connectivity between the default mode network and the striatum during reward anticipation*. Poster presented at the Society for Neuroeconomics. Boston, MA, USA.

Wyngaarden III, J.B., Nambiar A., Dennison J.B., Alloy L.B., Fareri D.S., Jarcho J.M., Smith D.V. (April 2025). *Trait reward sensitivity and motivation shape connectivity between the default mode network and the striatum during reward anticipation*. Poster presented at the Social & Affective Neuroscience Society. Chicago, IL, USA.

Dwamena D., **Wyngaarden III, J.B.**, Kos M., Sharp C.J., Yang J., Jarcho J.M., Fareri D.S., Smith D.V. (April 2025). *Dorsal anterior cingulate responses to unreciprocated trust are associated with neural responses to unfairness*. Poster presented at The Social & Affective Neuroscience Society. Chicago, IL, USA

Wyngaarden III, J.B., Sharp, C.J., Dachs, A., Sazhin, D., Tropea, T., Kohli, I., Giovannetti, T., Fareri, D.S., Jarcho, J.M., & Smith, D.V. (November 2024). *Risk for financial exploitation and psychosocial factors modulate age-related differences in social reward processing*. Poster presented at the Society for Neuroscience conference. Chicago, IL, USA.

Cunningham, R.M., Calderaro, M.L., Quarmley, M., Clarkson, T., Schmidt, H., **Wyngaarden III, J.B.**, Cassidy, C.M., & Jarcho, J.M. (September 2024). *Social anxiety in adolescents: Insights from neuromelanin-sensitive MRI in the substantia nigra*. Poster presented at the Flux Society conference. Baltimore, MD, USA.

- Wyngaarden III, J.B.**, Sharp, C.J., Dachs, A., Sazhin, D., Tropea, T., Kohli, I., Giovannetti, T., Fareri, D.S., Jarcho, J.M., & Smith, D.V. (April 2024). *Social support and risk for financial exploitation modulate age-related differences in reward processing*. Poster presented at the Social and Affective Neuroscience conference. Toronto, ON, CA.
- Wyngaarden III, J.B.**, Zaff, O., Sazhin, D., Dachs, A., Collins, M., Johnston, C.R., Giovannetti, T., Jarcho, J.M., & Smith, D.V. (November 2023). *Age-related differences in social reward processing*. Poster presented at the Society for Neuroscience conference. Washington DC, USA.
- Sazhin, D., **Wyngaarden III, J.B.**, Zaff, O., Ludwig, R., Collins, M., Sharp, C., Dachs, A., Nambiar, A., Dennison, J.B., Drayton, M., Tropea, T., Clithero, J., Giovannetti, T., & Smith, D.V. (November 2023). *Characterizing age-related differences in neural responses during the Ultimatum Game*. Poster presented at the Society for Neuroscience conference. Washington DC, USA.
- Yang, J., Zaff, O., **Wyngaarden III, J.B.**, Sazhin, D., Dachs, A., Sharp, C.J., Collins, M., Ludwig, R., Giovannetti, T., Fareri, D.S., & Smith, D.V. (November 2023). *Associations between age and neural response to reward in social contexts*. Poster presented at the Society for Neuroscience conference. Washington DC, USA.
- Dachs, A., Sharp, C.J., Yanilmaz, E., **Wyngaarden III, J.B.**, Zaff, O., Tropea, T., Sazhin, D., Ludwig, R., Giovannetti, T., Fareri, D.S., & Smith, D.V. (November 2023). *Examining age-related differences in neural response to trust*. Poster presented at the Society for Neuroscience conference. Washington DC, USA.
- Johnston, C.R.*, de Arruda, T.C.M.*, Todorovski, I., **Wyngaarden III, J.B.**, Smith, D.V., Jarcho, J.M. (September 2023). *Childhood maltreatment and memory bias for social and non-social events: exploring neural mechanisms that promote risk for mental health problems*. Poster presented at the Flux Society conference. Santa Rosa, CA, USA.
- Wyngaarden III, J.B.**, Sharp, C.J., Dennison, J.B., Zaff, O., Olson, I., Fareri, D.S., Jarcho, J.M., Smith, D.V. (July 2023). *Task-dependent functional connectivity with the cerebellum: Confounded by Nyquist ghosting?* Poster presented at the Organization for Human Brain Mapping conference. Montreal, QC, CA.
- Wyngaarden III, J.B.**, Johnston, C.R., Sazhin, D., Dennison, J.B., Vafiadis, A., Zaff, O., Kahn, N.A., Collins, M., Nguyen, A., Fareri, D., McCloskey, M., Alloy, L.B., Smith, D.V., & Jarcho, J.M. (March 2023). *Substance use is associated with reduced temporal-parietal junction activation and corticostriatal connectivity for social relative to monetary reward*. Poster presented at the Philadelphia Chapter of the Society for Neuroscience conference. Philadelphia, PA, USA.
- Wyngaarden III, J.B.**, Johnston, C.R., Sazhin, D., Dennison, J.B., Vafiadis, A., Zaff, O., Kahn, N.A., Collins, M., Nguyen, A., Fareri, D., McCloskey, M., Alloy, L.B., Smith, D.V., & Jarcho, J.M. (March 2022). *Alcohol use is associated with enhanced corticostriatal connectivity with precuneus during social reward*. Poster presented at the Social and Affective Neuroscience Society conference. Online.
- Wyngaarden III, J.B.**, Monroe, A.E., & Plant, A.E. (February 2020). *Moral values of fairness and authority play opposing roles in explaining people's support for racial protests*. Poster presented at the Society for Personality and Social Psychology conference. New Orleans, LA, USA.

Wyngaarden III, J.B. (February 2020). *Naming and morality: How having a name influences our perceptions of mind and moral status*. Talk given at the Three Minute Thesis (3MT) Council of Southern Graduate Schools competition in Birmingham, AL.

Wyngaarden III, J.B. (October 2019). *Naming and morality: How having a name influences our perceptions of mind and moral status*. Talk given at the Three Minute Thesis (3MT) competition at Appalachian State University.

Wyngaarden III, J.B., Monroe, A.E., & Plant, A.E. (October 2019). *Moral values of fairness and authority play opposing roles in explaining people's support for racial protests*. Talk given at the Society of Southeastern Social Psychologists conference. Johnson City, TN, USA.

RESEARCH EXPERIENCE

Postdoctoral Fellow *Department of Psychiatry, Icahn School of Medicine at Mount Sinai*
August 2026-Present

Responsibilities:

Postdoctoral research in the laboratory of Muhammad A. Parvaz, Ph.D., on drug cue reactivity and reward learning in stimulant use disorders. Research program applies latent growth curve modeling and reinforcement learning models to longitudinal EEG data from treatment-seeking individuals with cocaine and methamphetamine use disorders, and develops individualized neuromodulation protocols targeting EEG biomarkers of cue reactivity. Affiliated with NIDA-funded studies R37DA058039 and R33DA056779.

Doctoral Student *Department of Psychology and Neuroscience, Temple University*
August 2020-July 2026

Responsibilities:

Conducted independent research on decision-making and social reward processing across the lifespan, with emphasis on cognitive biases and financial exploitation in older adults. Led multimodal studies combining fMRI, behavioral paradigms, and transcranial electrical stimulation to investigate reward-based learning mechanisms. Dissertation project examined whether transcranial stimulation enhances reward learning in older adults by modulating prefrontal cortex function—addressing critical vulnerabilities including confirmation bias and financial exploitation susceptibility. Managed complex datasets using computational modeling (Win-Stay-Lose-Shift, Rescorla-Wagner), neuroimaging analysis (FreeSurfer, FSL), and programming in bash, Python, MATLAB, and R. Mentored undergraduate researchers and collaborated across multiple institutions. Supported by NIH F31 predoctoral fellowship (\$106,980) and SRNDNA pilot grant (\$24,000).

Master's Student *Department of Psychology, Appalachian State University*
August 2018-May 2020

Responsibilities:

Investigated social cognition and moral decision-making through experimental design and statistical analysis. Led thesis project examining mind perception and moral judgments. Managed full research pipeline from grant writing to publication. Trained undergraduate research assistants and presented findings at national conferences.

Research Assistant *Department of Psychology, New York University*
Summer 2019

Responsibilities:

Conducted behavioral and neuroimaging research on developmental differences in risk-taking and reward-memory interactions. Managed participant recruitment from large NYC databases and executed experimental protocols across age groups (8-25 years). Operated fMRI equipment and performed quality control analyses using MRIQC and fMRI Prep. Handled complex logistics of working with minors and families in clinical research settings.

Research Assistant *Department of Psychology, University of North Carolina at Charlotte*
August 2017-December 2017

Responsibilities:

Supported EEG research investigating concussion effects on cognitive control and mind-wandering. Assisted with low-impedance EEG data collection and experimental protocol execution.

Research Assistant *Department of Psychology and Neuroscience, University of North Carolina at Chapel Hill*
May 2015-May 2016

Responsibilities:

Investigated stress effects on goal-directed versus habitual decision-making using behavioral paradigms and physiological measures. Collected multi-modal data including heart rate, salivary cortisol, and behavioral responses in simulated fMRI environments. Performed wet lab techniques including hormone assays and data analysis using Excel macros.

TEACHING EXPERIENCE

Instructor of Record, Temple University

PSY2502 Foundations of Cognitive Neuroscience
Summer 2022 & Summer 2023

Teaching Assistant, Temple University

PSY2104 Sensation & Perception
Fall 2023

NSCI0817 Brain Matters
Spring 2022-Fall 2022

PSY2101 Foundations in Cognitive Psychology
Spring 2021

Teaching Assistant, Appalachian State University

PSY4213 Social Psychology Lab
Fall 2019-Spring 2020

PSY3213 Social Psychology
Fall 2018-Spring 2019

PSY4655 Contemporary Issues in Moral Psychology
Fall 2018-Spring 2019

SERVICE

Temple University Graduate Students Association (TUGSA)

Temple University

August 2020-2026

Responsibilities: TUGSA is the labor union for teaching assistants, research assistances, and graduate assistants at Temple University. Members of the graduate student union work to prepare for contract negotiations, organization and outreach events, and other work-related issues.

Graduate Ambassador

Cratis D. Williams School of Graduate Studies

Appalachian State University

Fall 2019-Spring 2020

Responsibilities: Graduate student ambassadors are selected to help lead and facilitate events for new and prospective graduate students. These events include campus tours, open houses, graduate fairs, and orientations.

Trails to Success 360° Summit Institute

Cratis D. Williams School of Graduate Studies

Appalachian State University

Fall 2019-Spring 2020

Responsibilities: The Trails to Success 360° Summit Institute is a professional and personal development program for a diverse group of students from various programs within the graduate school. The institute hosts discussion-based meetings every other week and covers topics including leadership development, research skills, and presentation skills.

PROFESSIONAL AFFILIATIONS

2025-2026 Society of Biological Psychiatry

2022-2025 Society for Neuroscience

2021-2025 Social and Affective Neuroscience Society

2022-2023 Organization for Human Brain Mapping

2019-2020 Association for Psychological Science

2019-2020 Society for Personality and Social Psychologists